

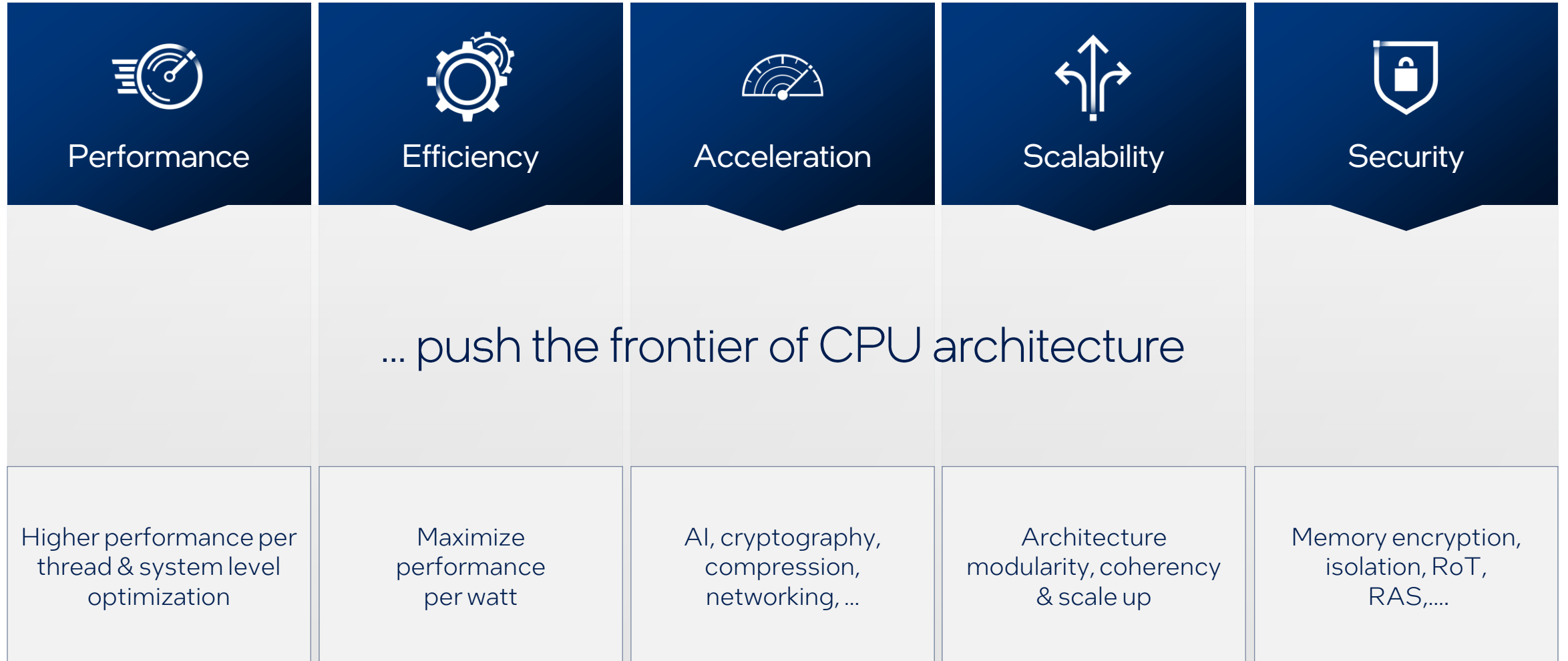


# Diamond Rapids

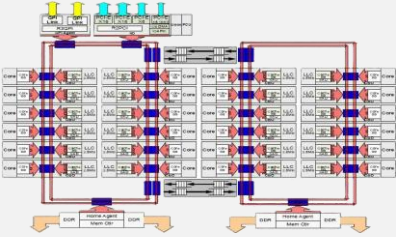
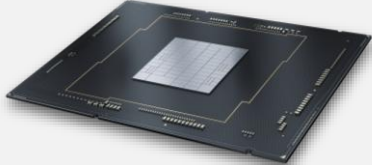
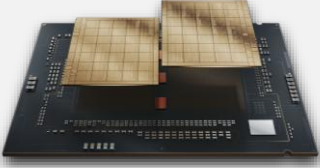
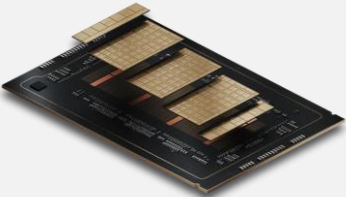
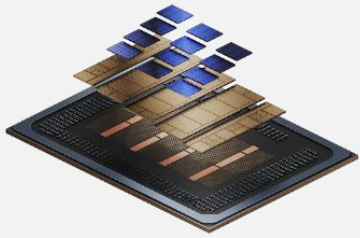
Intel's next generation Xeon CPU

Krishnakanth Sistla - Intel Fellow | Chief Architect Xeon Performance

# Evolving Data Center Workloads ....

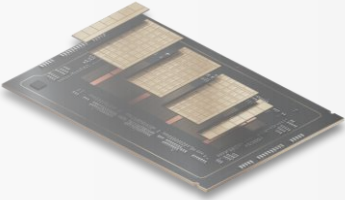
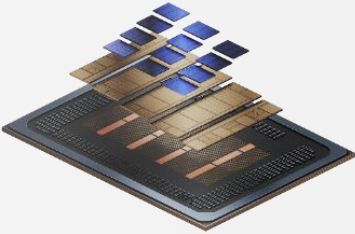


# Intel® Xeon® Architecture Journey

| Architecture<br><br>Packaging<br><br>Partitioning<br><br>Products<br><br>Brand | Ring                                                                              | Mesh                                                                                     | Tiled Mesh                                                                          | Modular Mesh                                                                        | 3D Mesh                                                                             |
|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                                                                                |  |         |  |  |  |
|                                                                                | Monolithic                                                                        | Monolithic                                                                               | Disaggregated 2.5D                                                                  | Disaggregated 2.5D                                                                  | 2.5D & 3D                                                                           |
|                                                                                | Centralized                                                                       | Centralized                                                                              | Tile Based Partition                                                                | Function Based Partition                                                            | Function Based Partition                                                            |
|                                                                                | <i>Haswell,<br/>Broadwell, ...</i>                                                | <i>Skylake, Cascade Lake,<br/>Cooper Lake, Ice Lake</i>                                  | <i>Sapphire Rapids,<br/>Emerald Rapids</i>                                          | <i>Granite Rapids,<br/>Sierra Forest</i>                                            | <i>Clearwater<br/>Forest</i>                                                        |
|                                                                                | Intel® Xeon® processors                                                           | 1 <sup>st</sup> , 2 <sup>nd</sup> , 3 <sup>rd</sup> Gen Intel® Xeon® Scalable processors | 4 <sup>th</sup> & 5 <sup>th</sup> Gen Intel® Xeon® Scalable processors              | Intel® Xeon® 6 with P-cores and E-cores                                             | Intel® Xeon® 6+ processors                                                          |

# The Next Frontier of DC CPU Architecture

Defined by how effectively the processor moves, processes and protects data at hyperscale

|                | Tiled Mesh                                                                                     | Modular Mesh                                                                      | 3D Mesh                                                                             |
|----------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                |               |  |  |
|                | Disaggregated 2.5D                                                                             | Disaggregated 2.5D                                                                | 2.5D & 3D                                                                           |
|                | Tile Based Partition                                                                           | Function Based Partition                                                          | Function Based Partition                                                            |
|                | <i>Sapphire Rapids, Emerald Rapids</i>                                                         | <i>Granite Rapids, Sierra Forest</i>                                              | <i>Clearwater Forest</i>                                                            |
| n <sup>®</sup> | 4 <sup>th</sup> & 5 <sup>th</sup> Gen Intel <sup>®</sup> Xeon <sup>®</sup> Scalable processors | Intel <sup>®</sup> Xeon <sup>®</sup> 6 with P-cores and E-cores                   | Intel <sup>®</sup> Xeon <sup>®</sup> 6+ processors                                  |

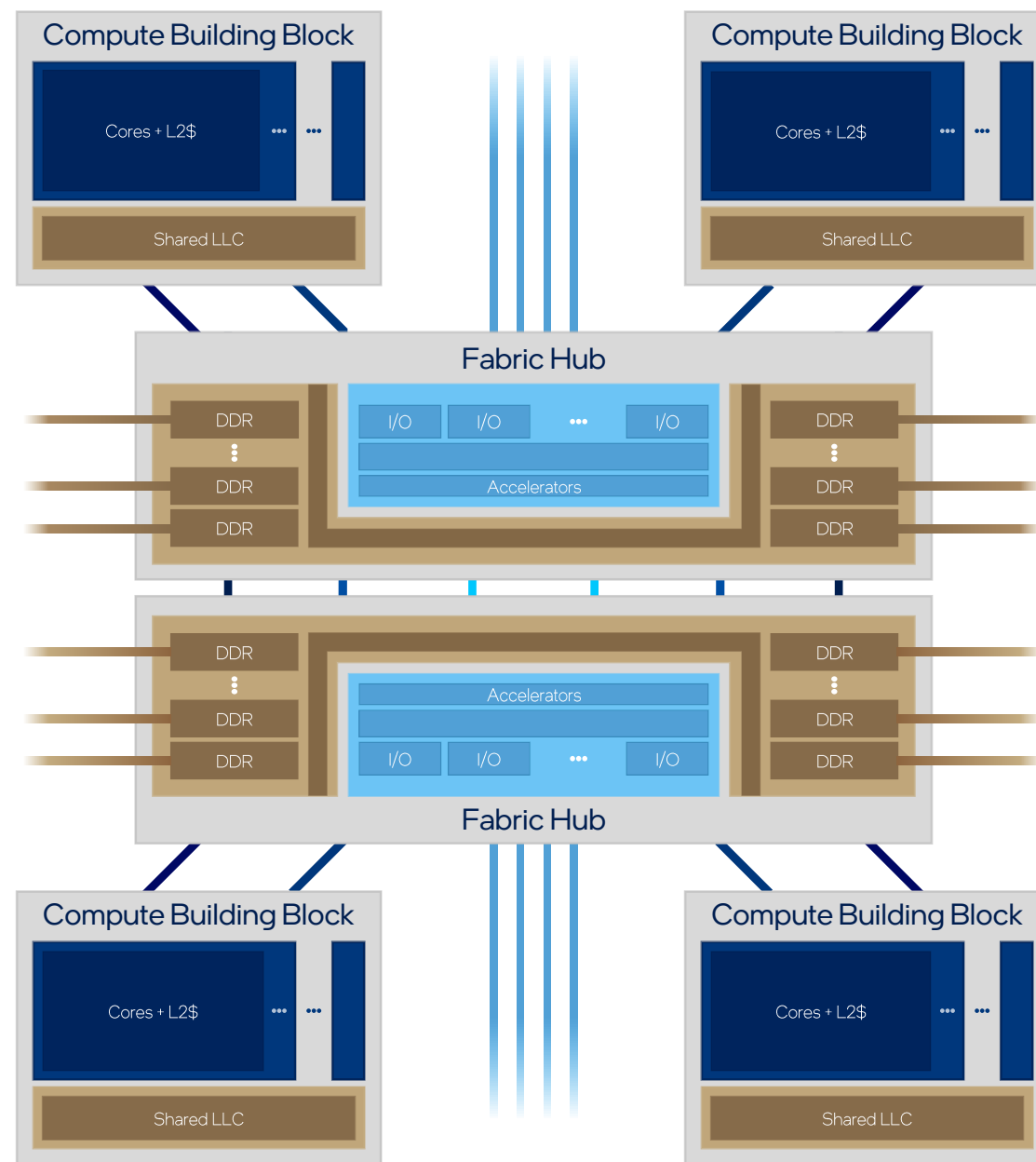
# Fan-out Fabric Architecture

Scalable Compute Building Blocks

Centralized I/O & Memory in Fabric Hub

Unified Memory Fabric

Flexible I/O Fabric



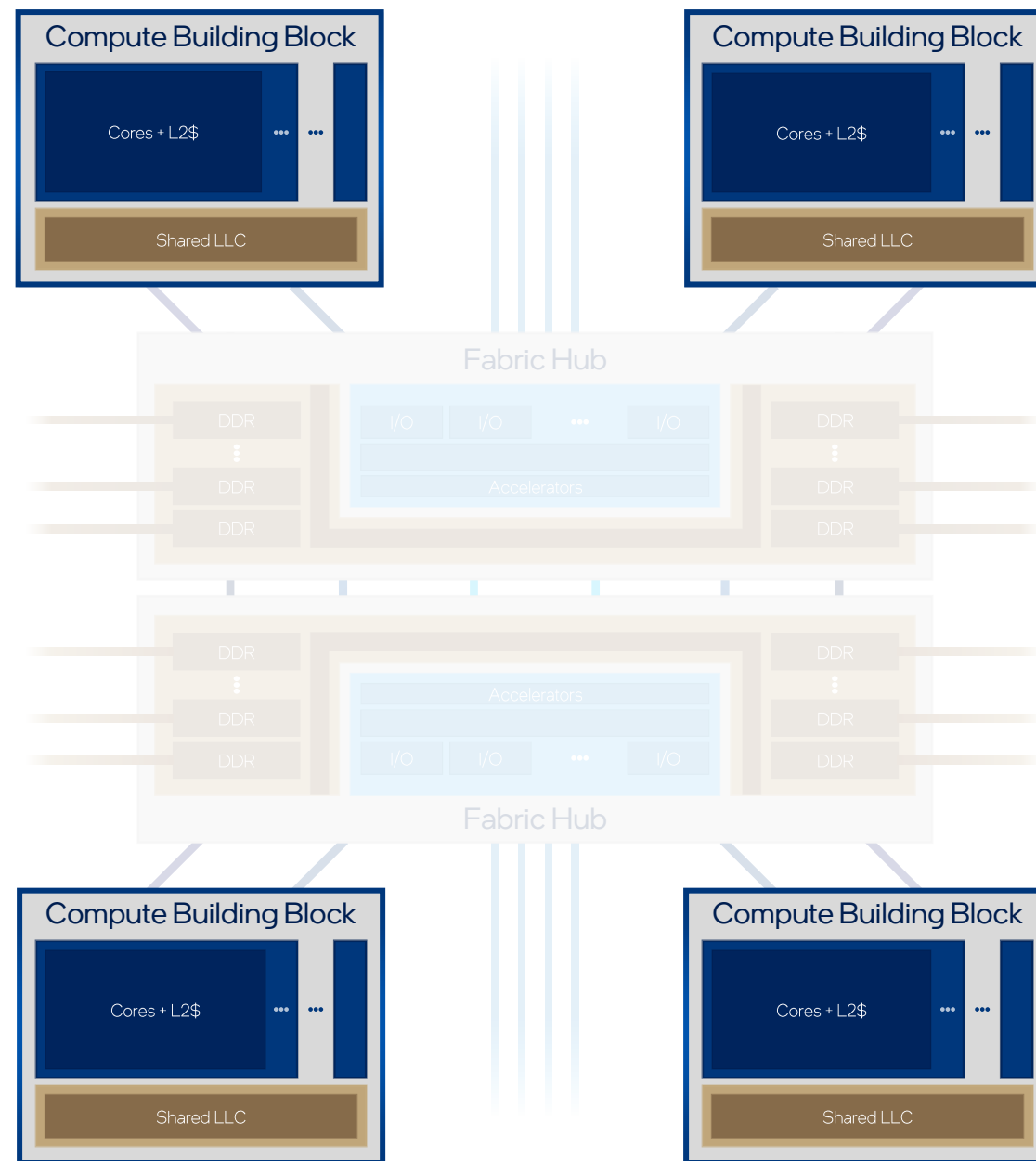
# Fan-out Fabric Architecture

## Scalable Compute Building Blocks

## Centralized I/O & Memory in Fabric Hub

## Unified Memory Fabric

## Flexible I/O Fabric



# Scalable Compute Building Blocks

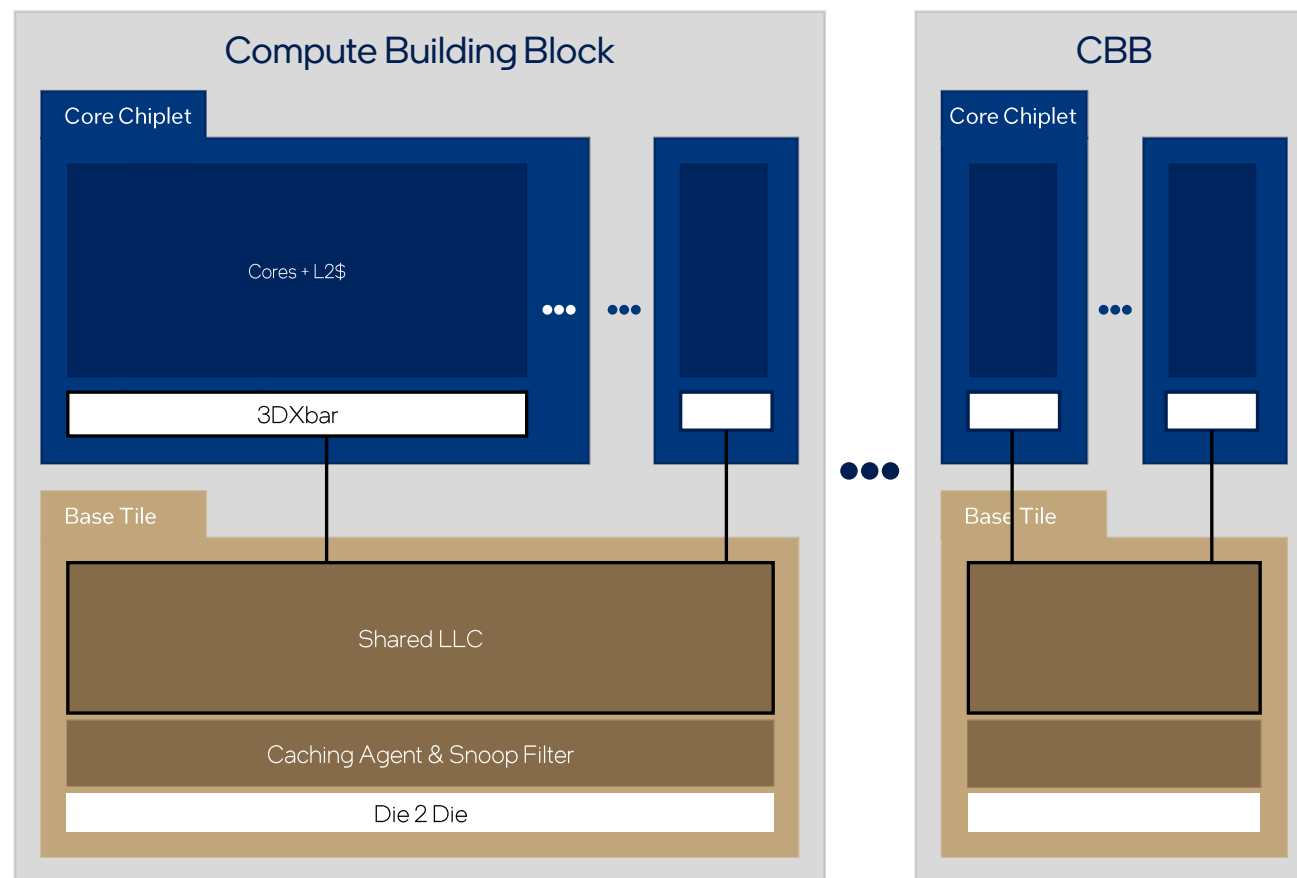
Up to 16 cores per **Core Chiplet**

**3D Xbar** connects cores to LLC on **Base Tile**

LLC in **Base Tile** shared across **Compute Building Block**

Scalable number of **core chiplets** per **Base Tile**

Scalable number of **Compute Building Blocks** / Chip



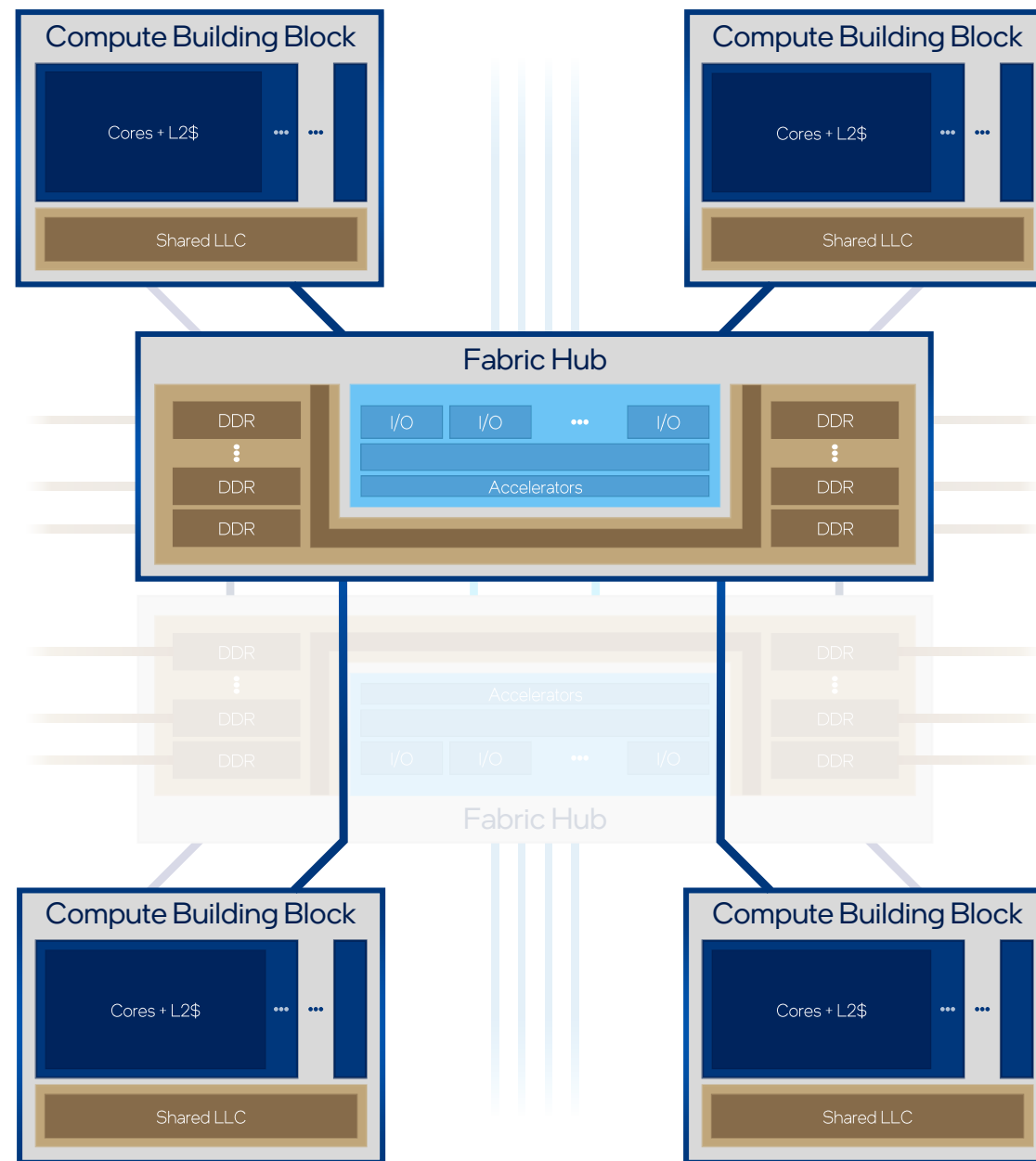
# Fan-out Fabric Architecture

Scalable Compute Building Blocks

Centralized I/O & Memory in Fabric Hub

Unified Memory Fabric

Flexible I/O Fabric





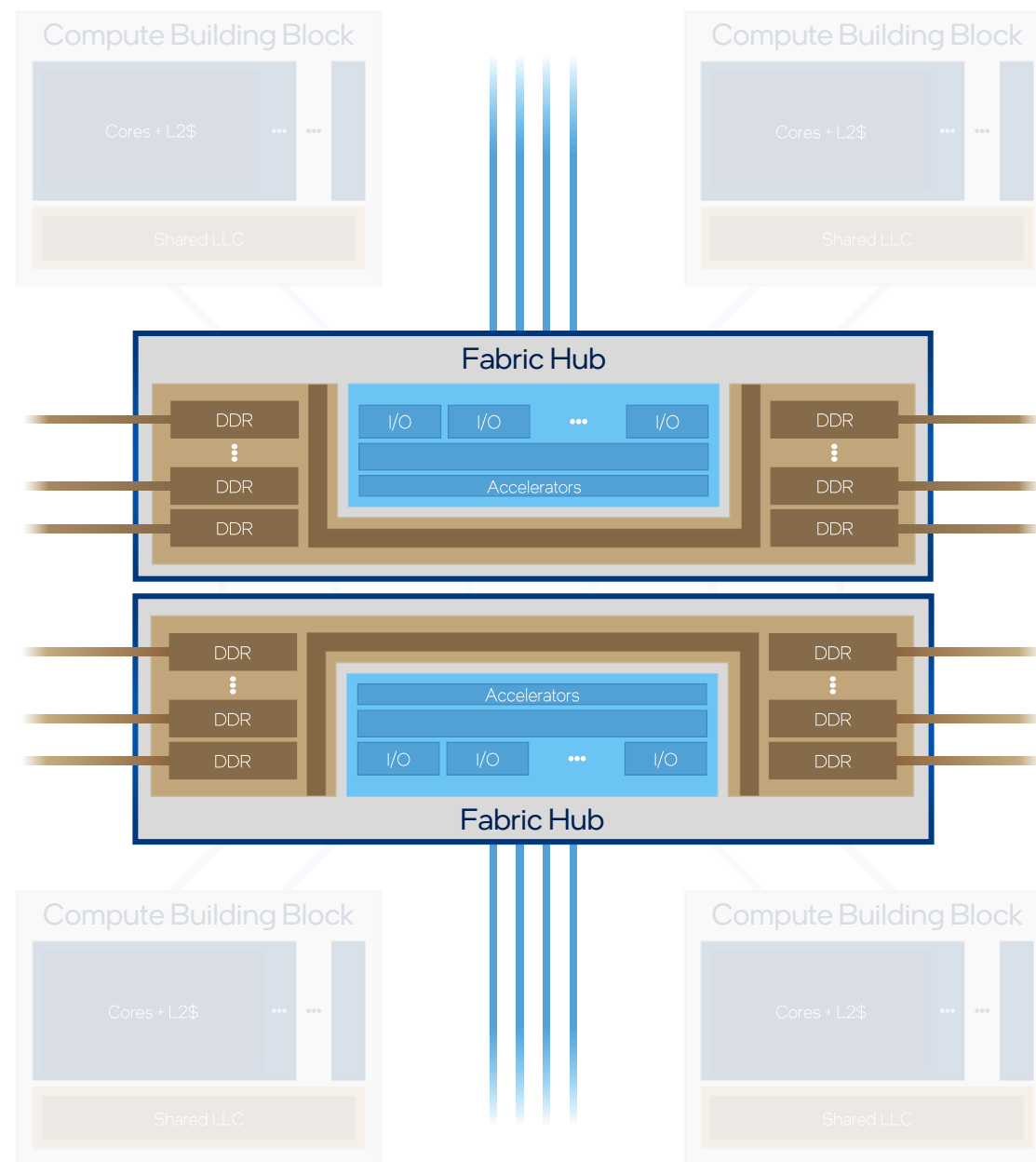
# Fan-out Fabric Architecture

Scalable Compute Building Blocks

Centralized I/O & Memory in Fabric Hub

Unified Memory Fabric

Flexible I/O Fabric



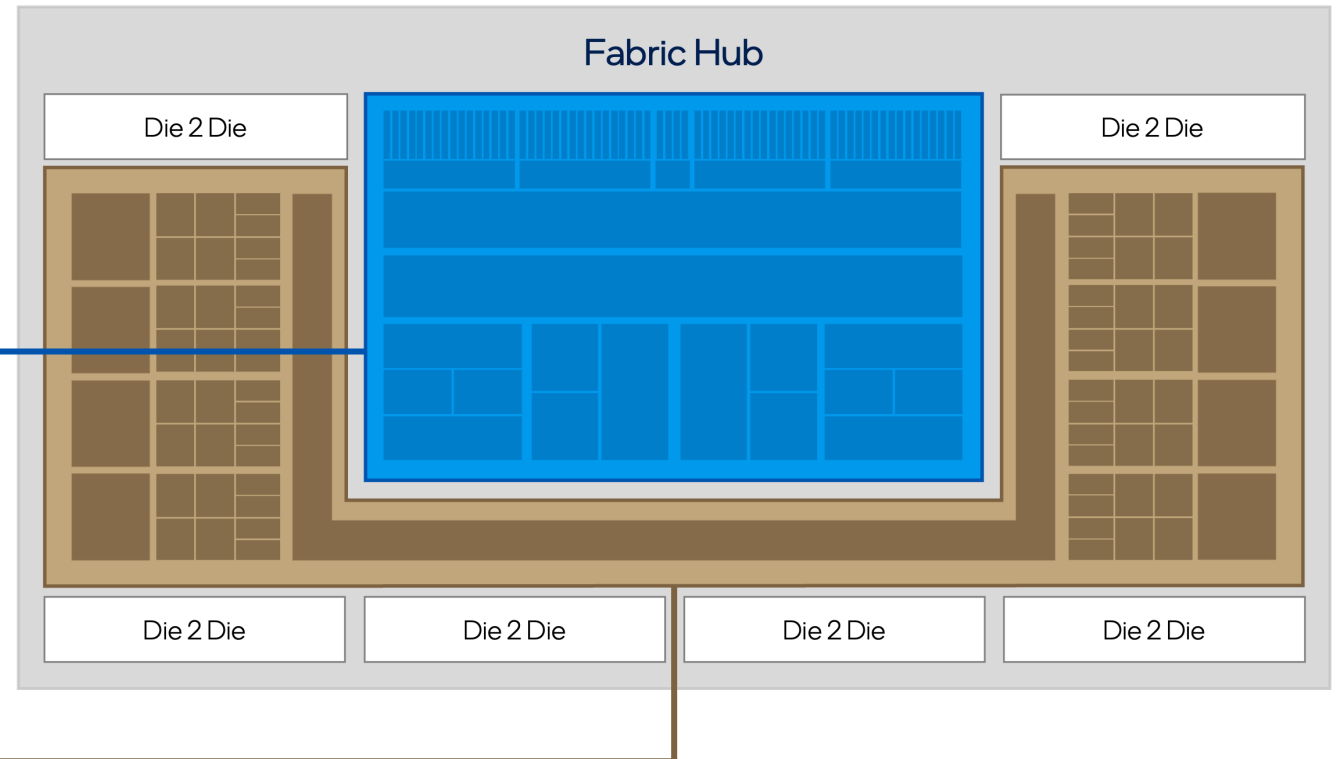
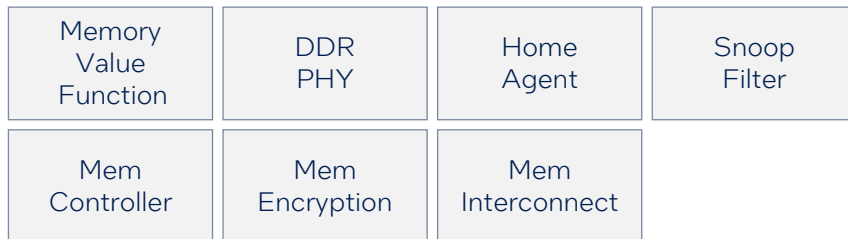
# Scalable Fabric Hub

## Flexbus I/O Fabric



## Unified Memory Fabric

Memory Subsystem with up to 1.6 TB/s bandwidth



# Unified Memory Fabric

## Memory Controller (MC)

schedules DDR accesses with robust ECC, power management, and row-hammer protection

## Memory Value Function (MVF)

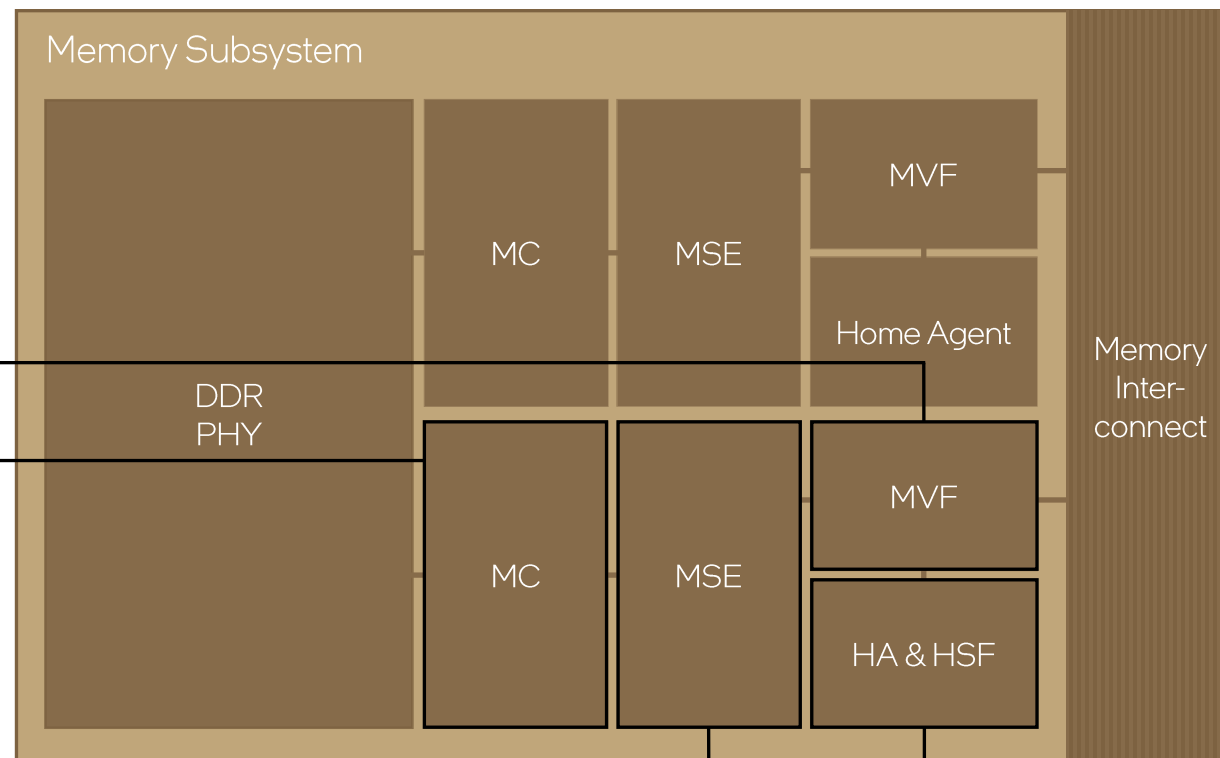
supports CXL memory (ILM or Flat2LM) and mirroring

## Memory Encryption Engine (MSE)

supports DDR or CXL memory encryption

## Home Agent (HA) & Home Snoop Filter (HSF)

On-Die Home Snoop Filter for coherency



# Flexbus I/O Fabric

## I/O PHY

4 x 16 high speed I/O lanes per Fabric Hub  
Additional 4 lanes of PCIe Gen4 for Platform Infra

## Flexible I/O Subsystem

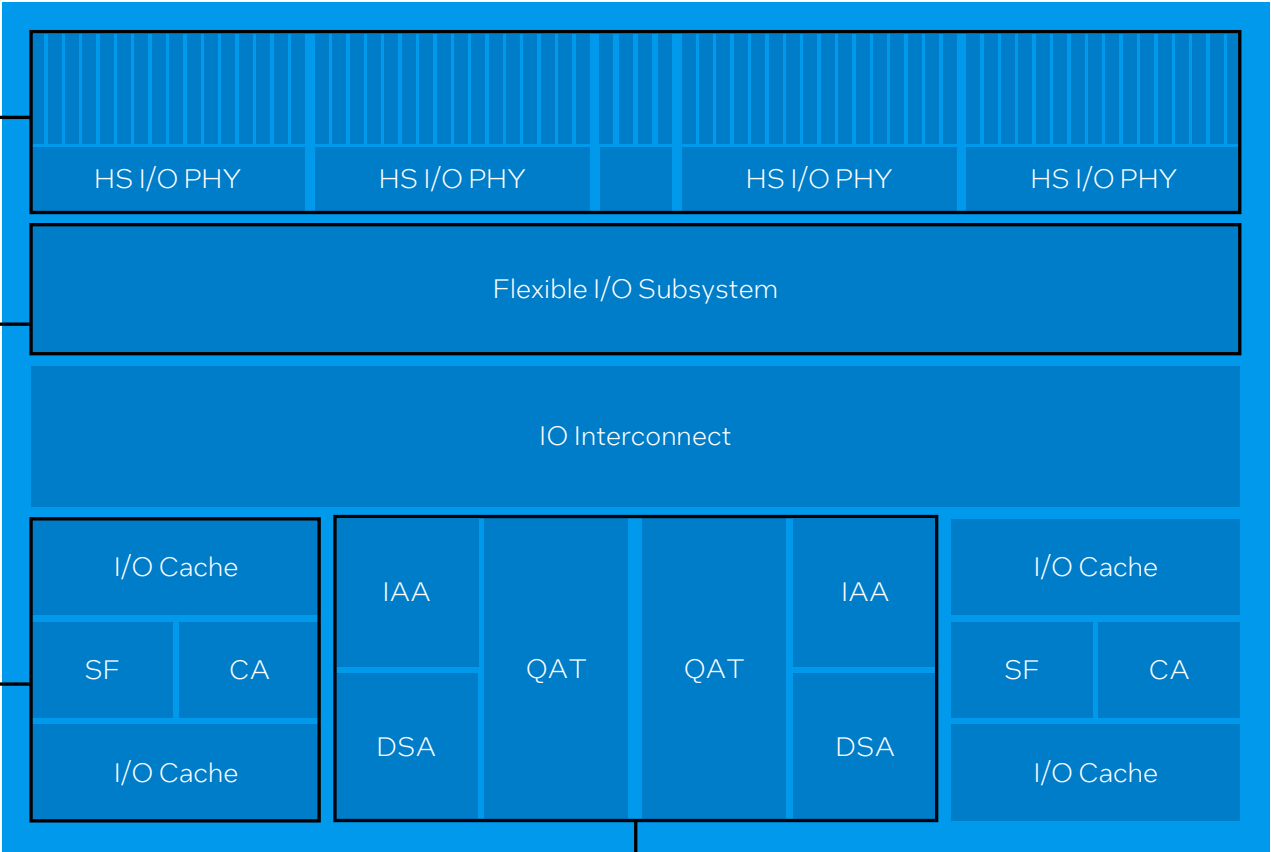
Each x16 can be used as PCIe Gen6, CXL 3, UPI 3 ports, or some combinations

## I/O Caching & Filtering

Up to 16MB of IO Cache, with on-die Snoop filtering and caching agent for coherency

## Accelerator Complexes

Each Fabric Hub houses 2 Accelerator complexes with QAT/DSA/IAA



# Power & Thermal Management

Distinct DVFS domains

## Core and CBB power management

New core idle power state with L2 cache retention

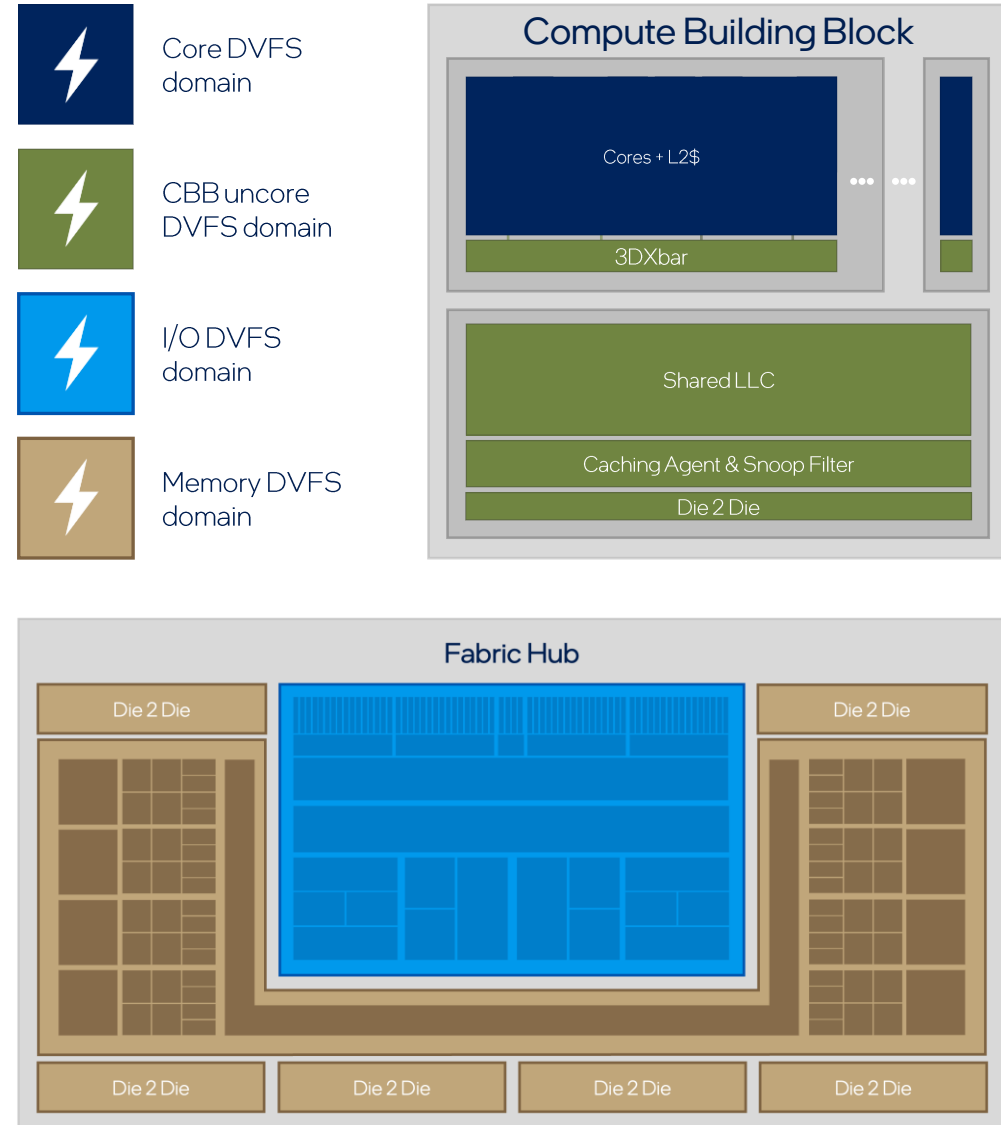
Priority Core Turbo support

## Fabric Hub power and thermal management

MR4 based memory thermal management

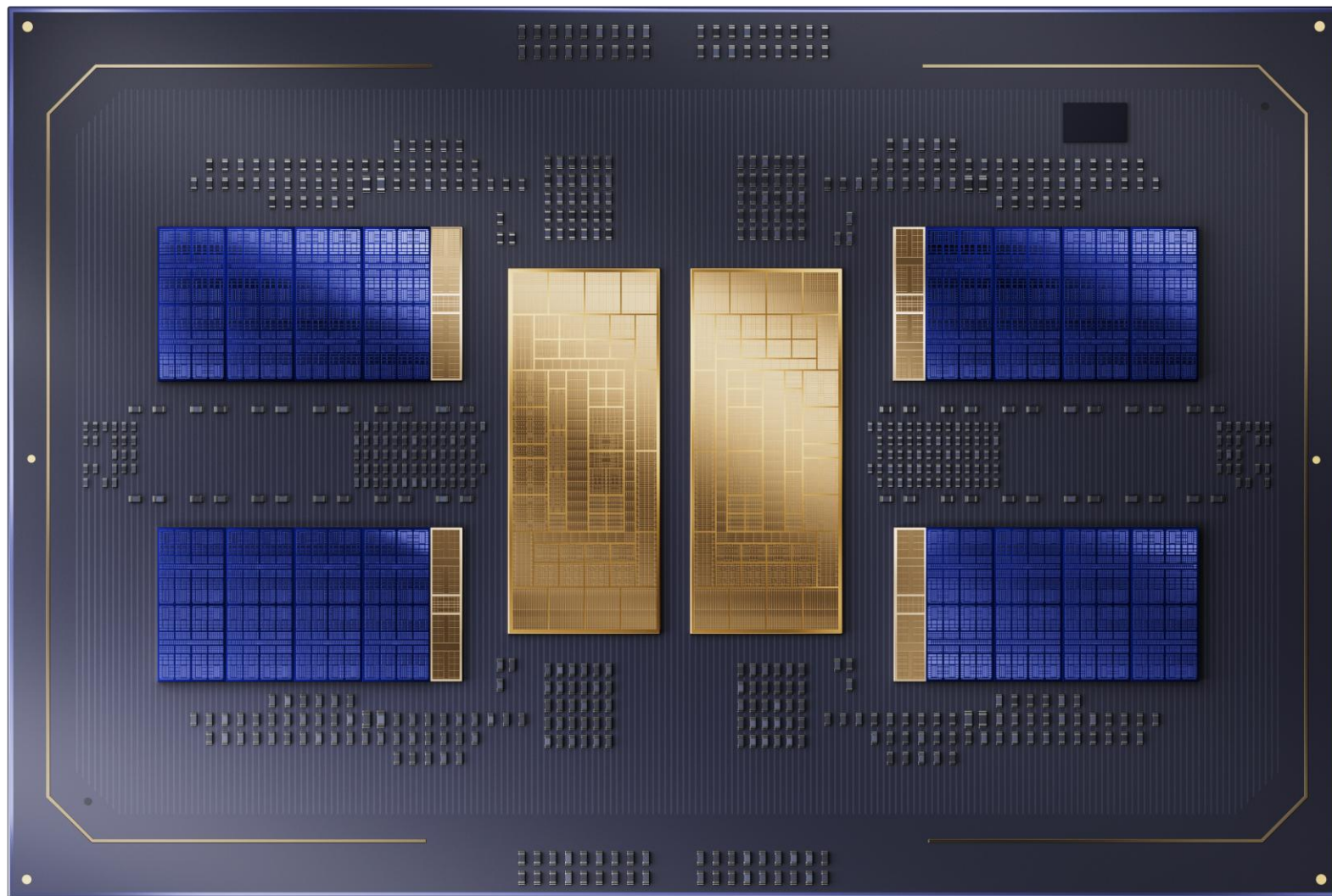
L0p support on both UXI and PCIe

Use-case based power direction to I/O & Mem



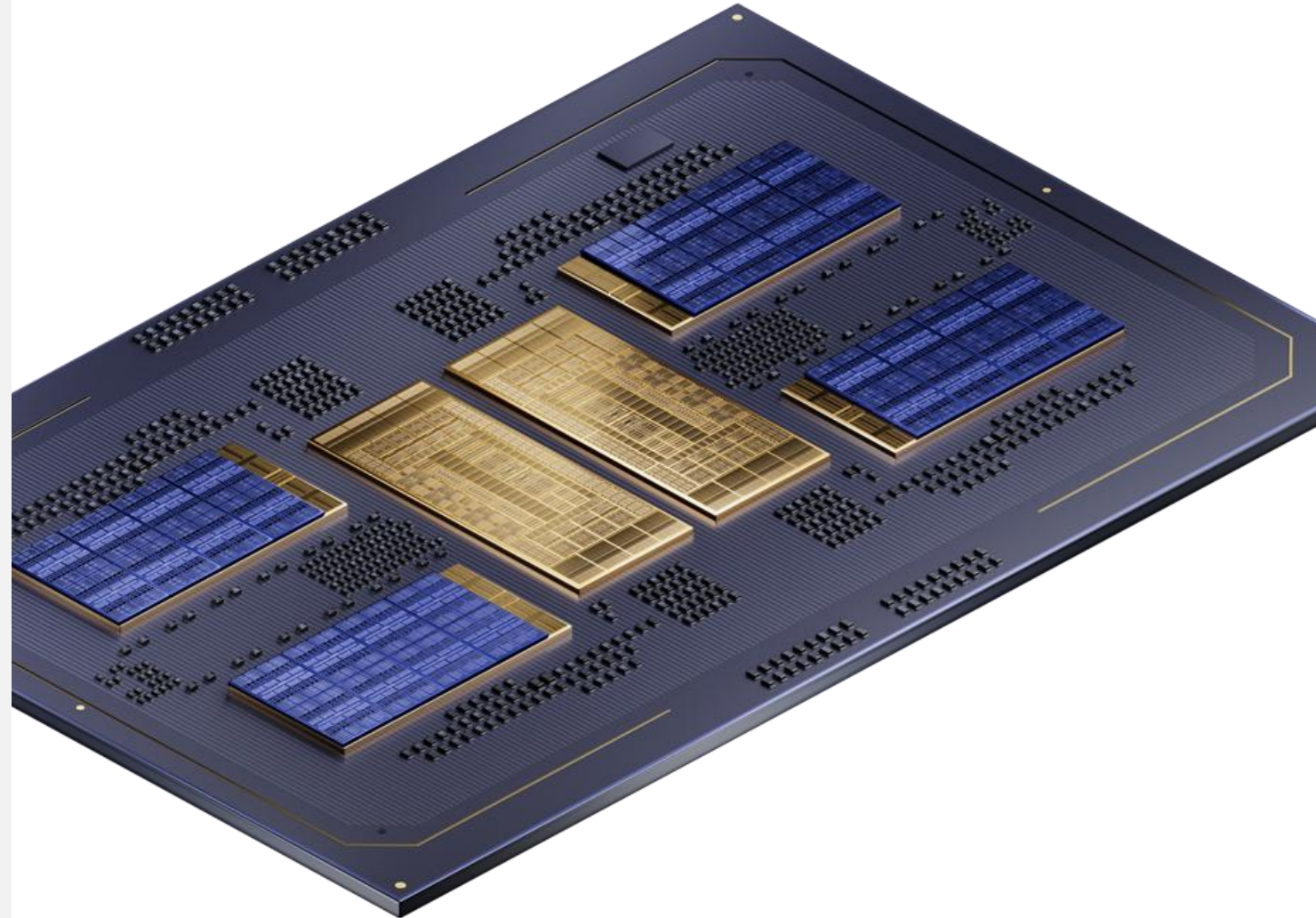
# Diamond Rapids

First Intel® Xeon® built  
with Fan-out Fabric  
Architecture



# Diamond Rapids SoC Overview

|                                    |                                         |
|------------------------------------|-----------------------------------------|
| <b>Compute</b>                     | 4 Compute Building Blocks               |
|                                    | Up to 1.28 GB of LLC across all CBBs    |
|                                    | Up to 256 cores total                   |
| <b>Memory</b>                      | 16 channels                             |
|                                    | Up to 8000MT/s DDR5                     |
|                                    | Up to 12800MT/s MRDIMM                  |
| <b>I/O</b>                         | 128 Lanes                               |
|                                    | PCIe Gen 6, CXL 3.0, UPI 3              |
|                                    | 8 lanes PCIe Gen 4                      |
| <b>Security &amp; Acceleration</b> | Intel TDX, Intel TDX connect, Intel SGX |
|                                    | Intel QAT, Intel DSA, Intel IAA         |
|                                    | Intel AMX, AVX-10                       |



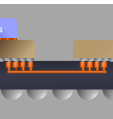


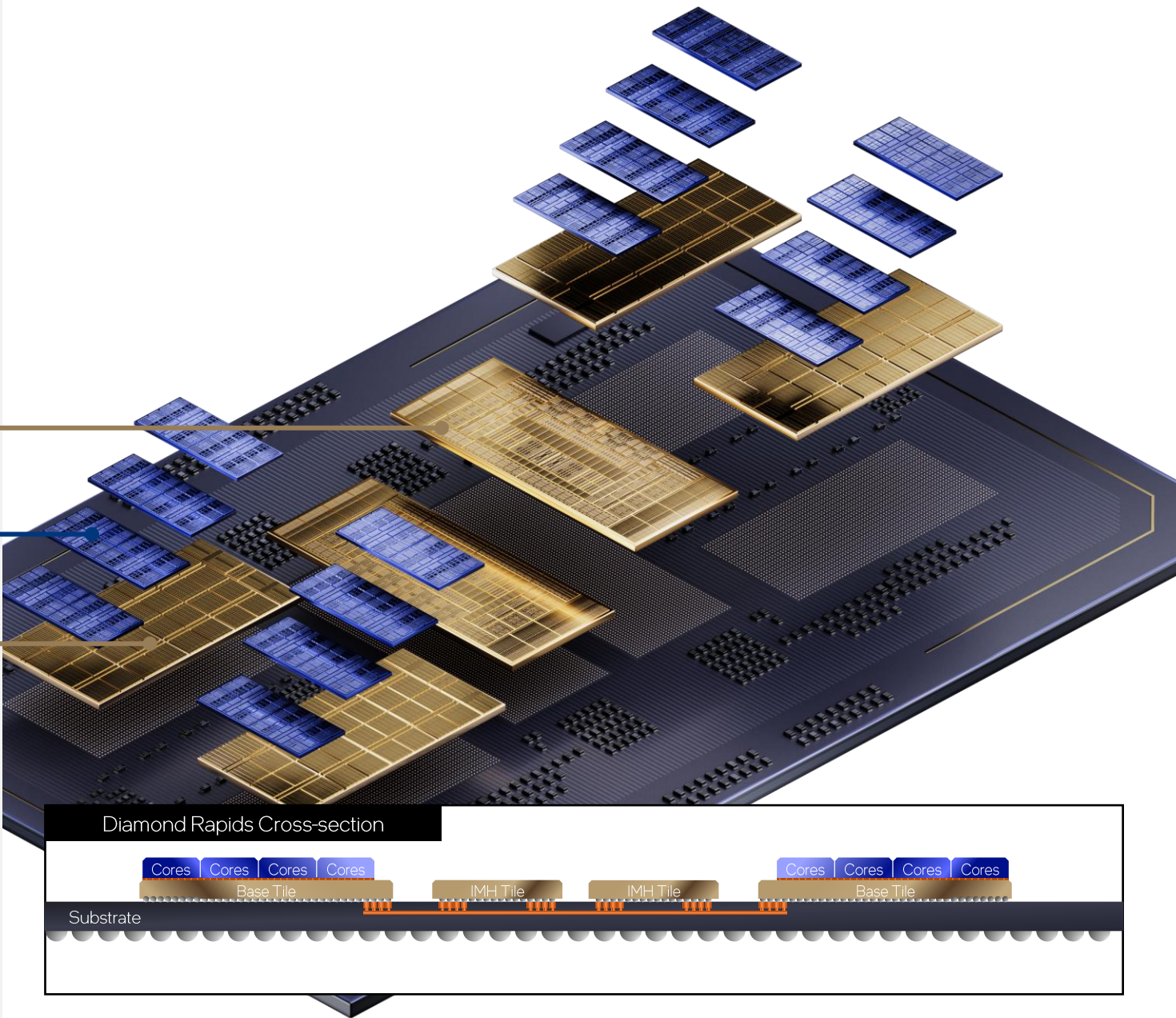
# Diamond Rapids Construction

## Process

|                    |             |
|--------------------|-------------|
| 2x Fabric Hub Tile | intel 3     |
| 16x Core Chiplets  | intel 18A-P |
| 4x Base Tile       | intel 3-T   |

## Packaging

|                                                         |                                                                                     |
|---------------------------------------------------------|-------------------------------------------------------------------------------------|
| Hybrid Bonding Interface (HBI)<br>Foveros 3D Direct     |  |
| Die to Die Interface<br>Substrate Copper Links (UCle-S) |  |







# Intel 18A-P

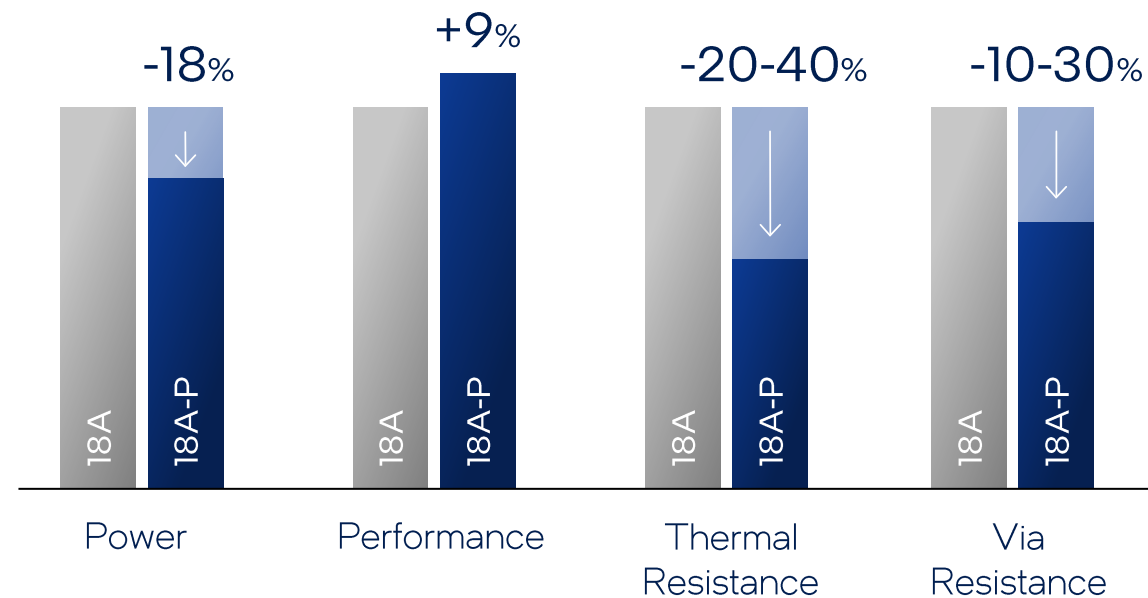
Expanded low-power, high-performance device offerings

PowerBoost dual-contact low-resistance option

New 5<sup>th</sup> logic Vt option between ULVT and LVT

33% tighter skew corners

Improved process variation



\*based on an industry standard core sub block

# Intel® Advanced Performance Extensions

Intel® APX

**Recompile without source code changes**

**Performance increase on recompiled SW**

**Legacy x86 code compatible**



# Security / RAS

## On-die snoop filter

replaces DRAM-resident directory state

## Preserves full ECC protection

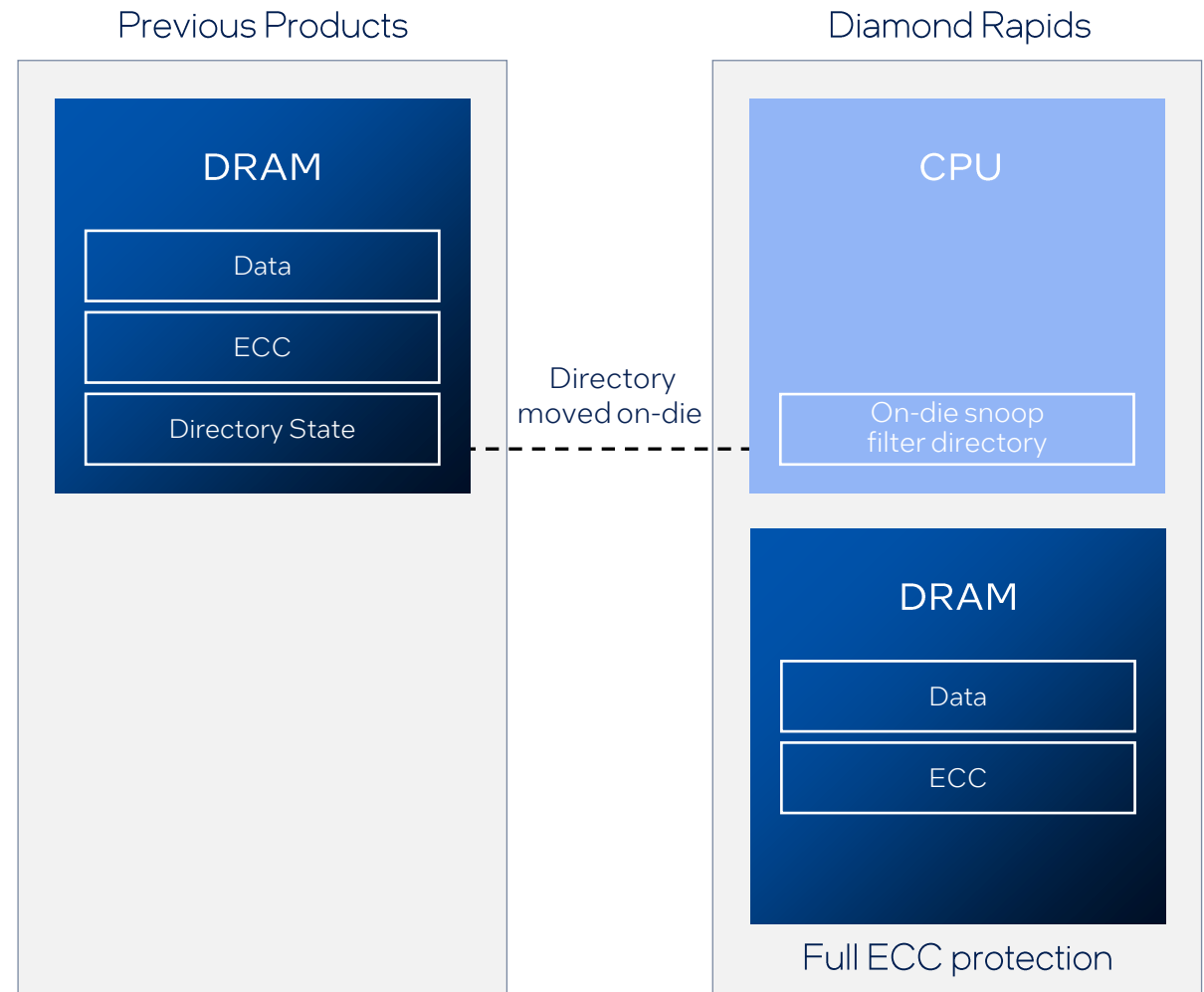
by eliminating directory storage in memory

## Lower latency & reduced coherence traffic

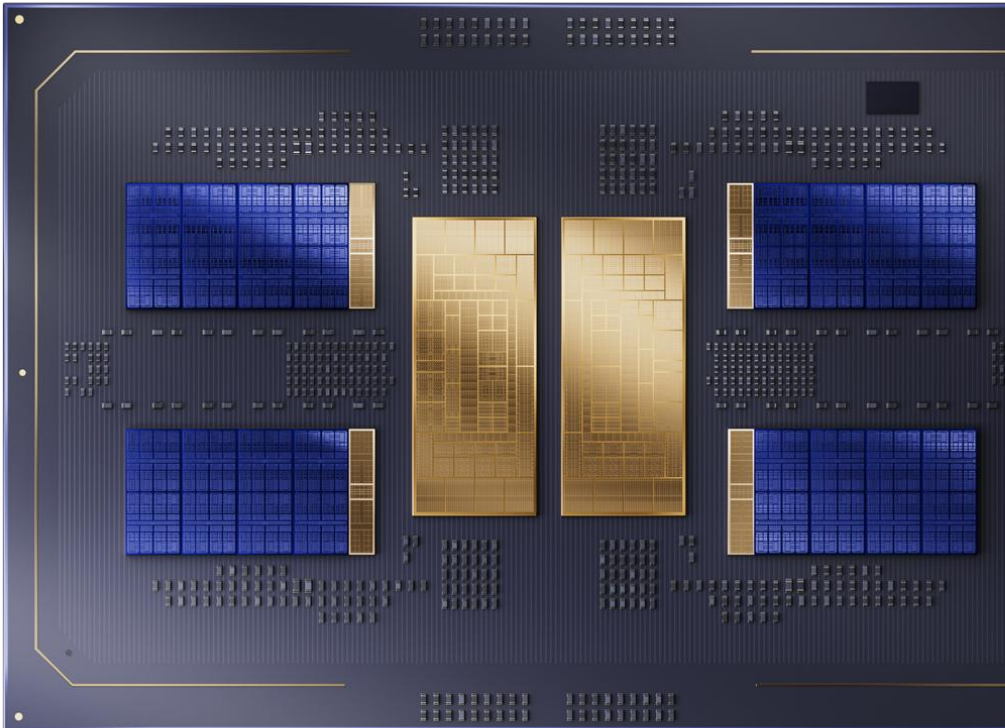
for cache lookups

## Simpler memory subsystem

with improved reliability



# Designed for Next Gen Agentic AI Infrastructure



High-performance core count

Up to  
**256**  
cores

Matrix multiplication acceleration &  
Enhanced vector instruction sets

**AMX & AVX 10.2**  
With FP8, FP16 & BF16 support

Large, fast, coherent cache

**1.28GB**  
Last Level Cache

High bandwidth memory support

**1.6TB/s**  
12800MT/s MRDIMM

High-Bandwidth, low latency I/O Fabric

**128 Lanes**  
up to 2TB/s Bi-directional  
bandwidth



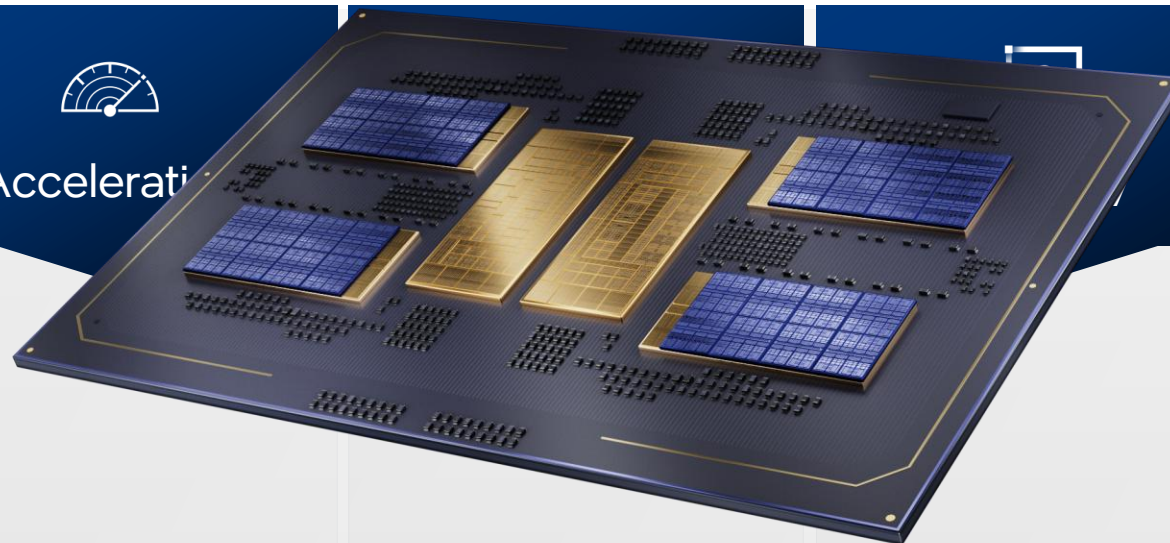
Performance



Efficiency



Acceleration



# Diamond Rapids

Intel's next generation Xeon CPU  
with performance cores



| High-throughput Compute | Efficient Performance  | Infrastructure & AI Acceleration  | Modular Architecture        | Multi Level Security     |
|-------------------------|------------------------|-----------------------------------|-----------------------------|--------------------------|
| Up to 256 cores         | Unified Memory Fabric  | Intel AMX                         | Fan-out Fabric Architecture | Intel TDX, Intel SGX     |
| Built-with 18A-P        | Independent DVFS       | AVX 10.2                          | Modular CBBs                | Improved RAS             |
| Massive LLC             | Lower Power I/O Fabric | Intel QAT / Intel DSA / Intel IAA | Foveros 3D Direct (HBI)     | Enhanced QoS & Telemetry |

# Notices & Disclaimers

Code names are used by Intel to identify products, technologies, or services that are in development and not publicly available.

These are not "commercial" names and not intended to function as trademarks.

Performance varies by use, configuration and other factors. Learn more at [intel.com/performanceindex](https://intel.com/performanceindex).

Performance results are based on testing as of dates shown in configurations and may not reflect all publicly available updates. See backup for configuration details. No product or component can be absolutely secure.

Your costs and results may vary.

Intel technologies may require enabled hardware, software or service activation.

© Intel Corporation. Intel, the Intel logo, and other Intel marks are trademarks of Intel Corporation or its subsidiaries. Other names and brands may be claimed as the property of others.





Thank You